

Smart Grid Investment Readiness Scorecard for Ontario Zones

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Abstract

This project investigates operational stress and infrastructure volatility across Ontario's electricity zones using real-world datasets from the Independent Electricity System Operator (IESO). By analyzing dispatch deviations, outage frequency, adequacy shortfalls, and zonal demand variability, a composite resilience score was developed to rank zones by empirical risk.

A total of **twelve experiments** were conducted, including time-series analysis, anomaly detection, forecast error comparison, and regression modeling. Key experiments synthesized multiple indicators to create a zone-level stress index. A linear regression model was trained to predict dispatch deviations using outage and adequacy metrics, with evaluation performed using both regression and classification metrics.

Results identified TORONTO, WEST, and ESSA as high-stress zones, supporting targeted infrastructure investment. The study offers value to researchers, policymakers, and consumers by highlighting zones most in need of smart grid modernization and by proposing a scalable methodology for resilience assessment.

Objective

To determine **which Ontario electricity zones exhibit the highest infrastructure stress and demand volatility**, based on empirical indicators such as:

- Real-time zonal demand
- Transmission outages
- Dispatch deviations
- Adequacy shortfalls
- Forecasting errors

The results aim to support smart grid planning, resilience investment prioritization, and operational risk assessment.

Experimentation and Analytical Workflow

The experiments ranged from time-series exploration, outlier detection, and variability scoring, to composite metric formulation and regression modeling. Data was preprocessed through extensive cleaning, transformation, and standardization.

Out of 12 designed experiments, the following were deemed most relevant:

Experiment	Purpose
Experiment 1 – Demand Volatility Analysis	Used Coefficient of Variation (CV) to rank zones by demand instability
Experiment 2 – Anomaly Detection	Applied Z-score and statistical thresholds to identify abnormal events by zone
Experiment 3 – Composite Stress Modeling	Synthesized deviation rates, outage frequency, and adequacy shortfalls
Experiment 6 – Forecast Error Comparison	Compared actual demand vs. VG and SBG forecasts
Experiment 10 – Forecast Uncertainty Analysis	Evaluated reliability of forecast mechanisms

The remaining experiments provided exploratory insights but were excluded from the final scoring and modeling framework due to either redundancy, data incompleteness, or marginal contribution.

Research Question 1:

Which Ontario electricity zones exhibit the highest volatility and infrastructure stress based on empirical indicators?

BRUCE, ESSA, and EAST showed the highest composite stress scores, indicating elevated risk from both unstable demand and frequent outages. These zones should be prioritized for smart grid upgrades, forecasting enhancements, or asset reinforcement.

Limitation

Experiment 1: No zone-level generation data; reserve margins estimated; time alignment was tricky.

This limits precision in assessing adequacy at the zonal level.

Experiment 2: Partial data; equal weights may misrepresent reality.

The composite score may not reflect the true operational importance of each metric.

Experiment 3: Oversimplifies grid stress; ignores local constraints and seasonality.

The Gap Score may overlook short-term or localized vulnerabilities.

Experiment 6: VG+SBG don't represent full supply; errors reflect structural gap, not forecast failure.

Results highlight limited VG/SBG coverage rather than inaccurate forecasting.

Experiment 10: Limited time range; forecast errors not zone-specific or causally isolated.

Forecast reliability is difficult to interpret without broader temporal and causal context.

Research Question: The analysis uses planned outages as a proxy for infrastructure stress and CV for volatility, which may distort low-demand zones. Equal weighting may oversimplify true impact, and the limited Jan–Jun 2025 timeframe could miss seasonal patterns. No external validation was used.

Summary of Technical Additions

To complement the experimentation and findings of this study, the following sections formalize the technical process used to evaluate Smart Grid readiness across Ontario zones. These additions provide detailed insights into data handling, analytical rigor, and modeling transparency.

Data Analysis Section

Summary of Key Dataset Features Using Descriptive Statistics

This project utilized operational datasets from Ontario's Independent Electricity System Operator (IESO), including real-time zonal demand, transmission outages, dispatch deviations,

and adequacy shortfall reports. Key statistical indicators—such as mean, standard deviation, and coefficient of variation (CV)—were computed for each zone to quantify demand volatility and service irregularities.

- Zonal CV highlighted instability in TORONTO and WEST.
- Outage logs revealed high-frequency events in ESSA and TORONTO.
- Dispatch deviations showed clustering in zones with high demand and low margin.

Visualizations Highlighting Patterns, Trends, or Outliers

- **Time-Series Trends:** Plotted for hourly demand and forecast residuals.
- **Heatmaps:** Used for zonal stress comparisons.
- **Boxplots:** Showed spread of actual vs. forecast demand.
- **Radar Charts:** Depicted composite stress rankings across zones.

These visualizations helped expose anomalies and zone-specific reliability issues.

Discussion of Potential Implications

- Persistent volatility in urban zones indicates potential undersupply or infrastructure bottlenecks.
- High deviation counts suggest operational constraints or real-time balancing issues.
- Forecasting errors (especially in VG and SBG) highlight a need for improved prediction models in planning processes.

Data Preparation Section

Explanation of Data Preprocessing Steps

Key preprocessing tasks included:

- **Cleaning:** Removal of null entries, malformed XML records, and irrelevant logs.
- **Scaling:** All metrics normalized between 0 and 1 for composite index computation.
- **Aggregation:** Monthly summaries were created from hourly demand and daily adequacy reports.
- **Filtering:** Focused on ten core Ontario electricity zones with consistent data coverage.

Documentation of Transformations

- Adequacy reports were parsed to extract daily reserve margins and identify shortfalls.
- Dispatch deviation reasons were grouped and summed monthly.
- Forecast error computed as $|\text{Actual} - \text{Forecast}|$ across SBG and VG datasets.
- Z-score anomaly tagging was applied to demand time series to flag events $>2.5\sigma$.

Model Evaluation Section

Step-by-Step Process of Model Development

- **Feature Selection:** OutageCount and AdequacyShortfall were used as inputs.
- **Target Variable:** TotalDeviations (monthly per zone)
- **Model Type:** Linear regression (scikit-learn)
- **Train-Test Split:** 70/30 ratio

Evaluation of Model Performance

- **Regression Results:**
 - ♦ MAE: 3.12
 - ♦ RMSE: 4.76
 - ♦ R^2 Score: 0.41
- **Binary Classification (via 60th percentile threshold):**
 - ♦ Accuracy: 73%
 - ♦ Precision: 68%
 - ♦ Recall: 71%
 - ♦ F1 Score: 0.695

Documentation of Findings and Insights

The model revealed moderate correlations between outages, adequacy issues, and deviation rates, but the limited R^2 suggests a need for more complex, perhaps non-linear, models. However, the **composite resilience score**, derived from normalized volatility, outage, and anomaly indicators, proved to be an effective comparative framework for zone-level investment readiness assessment.

Composite Resilience Score

A key innovation of this study is the construction of a **composite stress score** that combines:

1. Normalized demand volatility
2. Z-score-based anomaly counts
3. Outage frequency

Each metric was scaled between 0 and 1 and averaged. Zones were ranked based on this score to assess relative operational risk.

Top high-stress zones identified:

- TORONTO
- WEST
- ESSA

Research Perspectives

From a Researcher's Viewpoint:

The project successfully demonstrates that operational stress can be empirically captured using integrated metrics. While regression modeling revealed moderate predictability, the composite score proved effective for comparative analysis.

From a Government or Utility Perspective:

The zone ranking and resilience scoring can help prioritize infrastructure investments, predict stress periods, and guide grid modernization efforts in regions with high volatility and anomalies.

From a Consumer Perspective:

Residents and businesses in high-risk zones are more likely to experience power fluctuations, pricing anomalies, and outages. Insights from this analysis can inform consumer decisions around energy resilience (e.g., backup solutions) and advocacy for grid upgrades.

Conclusion

This study introduces a data-driven methodology to evaluate Smart Grid investment readiness across Ontario's electricity zones. Through twelve structured experiments and a composite resilience scoring framework, the research successfully highlights zones—particularly **TORONTO**, **WEST**, and **ESSA**—with elevated operational stress and infrastructure volatility.

The use of descriptive analytics, forecast validation, and linear modeling provided both interpretability and replicability. While the regression model established baseline predictive relationships, the composite index proved superior for practical zone-level ranking and policy prioritization.

These findings provide actionable insights for:

- **Policymakers:** to prioritize infrastructure upgrades.
- **Utilities:** to proactively manage grid stress and forecasting errors.
- **Researchers:** to explore advanced modeling techniques for grid resilience.

Future work will expand on this by integrating nonlinear models, broader datasets (weather, pricing, renewable generation), and deploying real-time dashboards for dynamic investment planning. This methodology can serve as a foundational decision-support tool in Ontario's path toward smart, resilient, and sustainable grid modernization.

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